

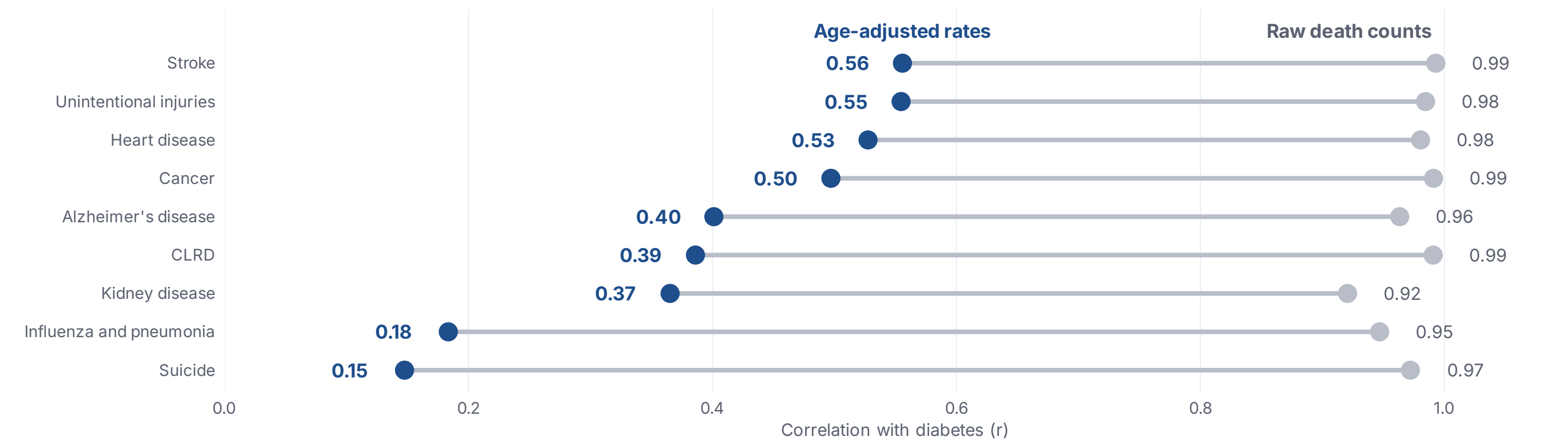
Diabetes and the other leading causes of death in the US

States with higher diabetes death rates also have higher stroke, heart disease and cancer death rates: a link you only see when you compare rates, not raw counts.

Akuete Giana MATHEY-APOSSAN · IST 719 Information Visualization · Syracuse University · Data: CDC / NCHS, Leading Causes of Death, 1999–2017 (10,868 rows)

1. Why counts mislead

Pearson correlation with diabetes across 51 states (50 + DC), 1999–2017.
Big states have more deaths from everything, so counts 'correlate' with any cause.

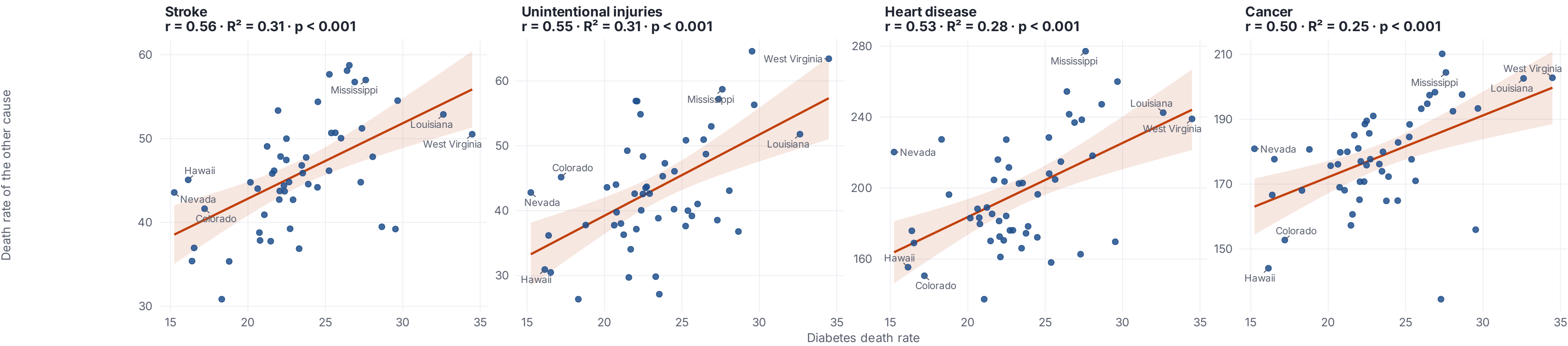


Key findings

- Raw counts give $r = 0.92\text{--}0.99$ for every cause: a population artefact.
- On rates, diabetes tracks stroke ($r = 0.56$), unintentional injuries (0.55) and heart disease (0.53).
- Deaths rose 22% but the age-adjusted rate fell 14% (1999–2017).
- Diabetes is the 7th leading cause of death (2017).

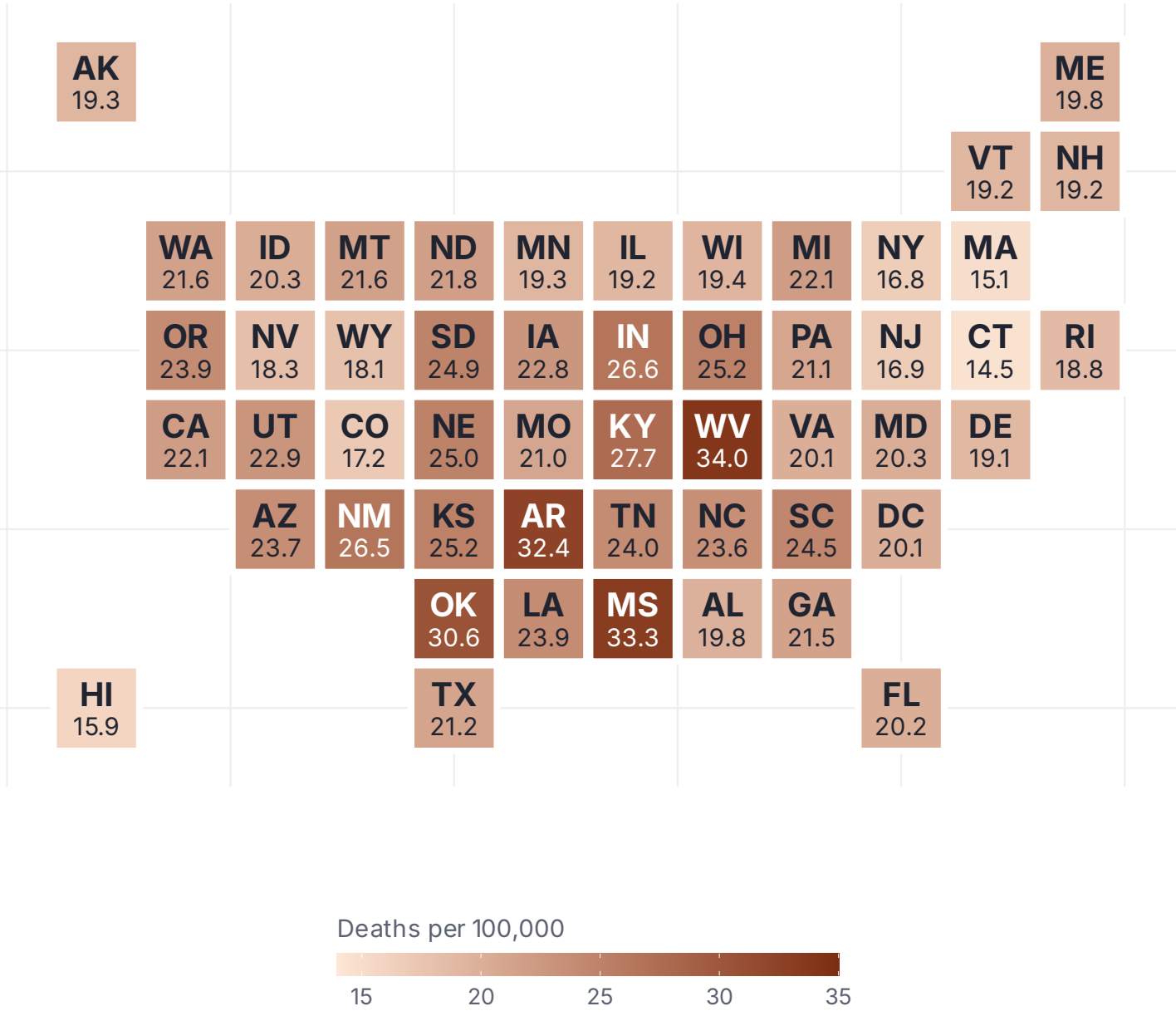
2. On rates, diabetes travels with stroke, injuries, heart disease and cancer

Each dot is a state: age-adjusted deaths per 100,000, averaged 1999–2017. Shared risk factors (obesity, smoking, inactivity, poverty, rurality, access to care) likely link them. This is correlation, not causation.



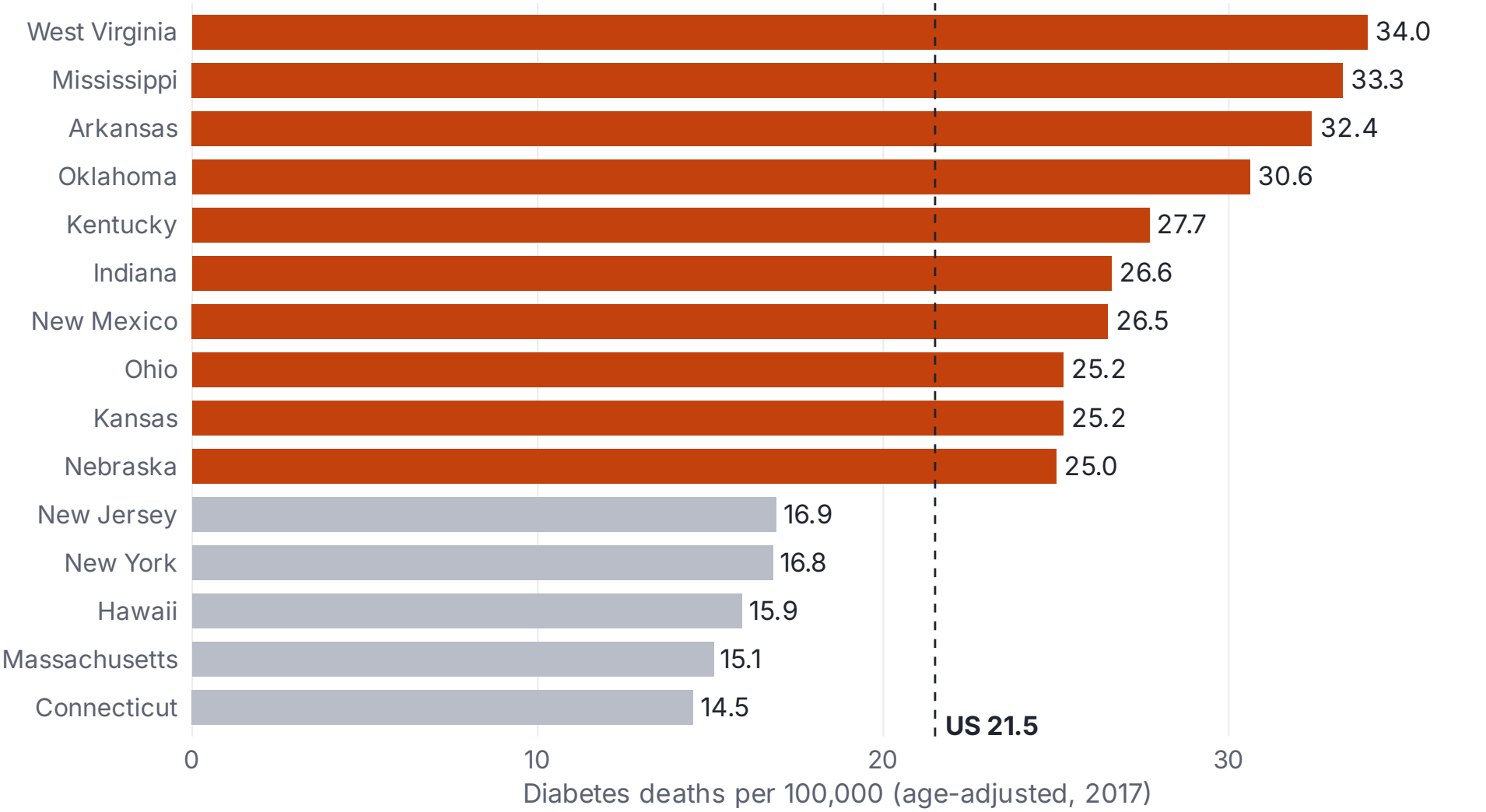
3. Where diabetes deaths are highest (2017)

Age-adjusted diabetes death rate per 100,000. US average: 21.5.



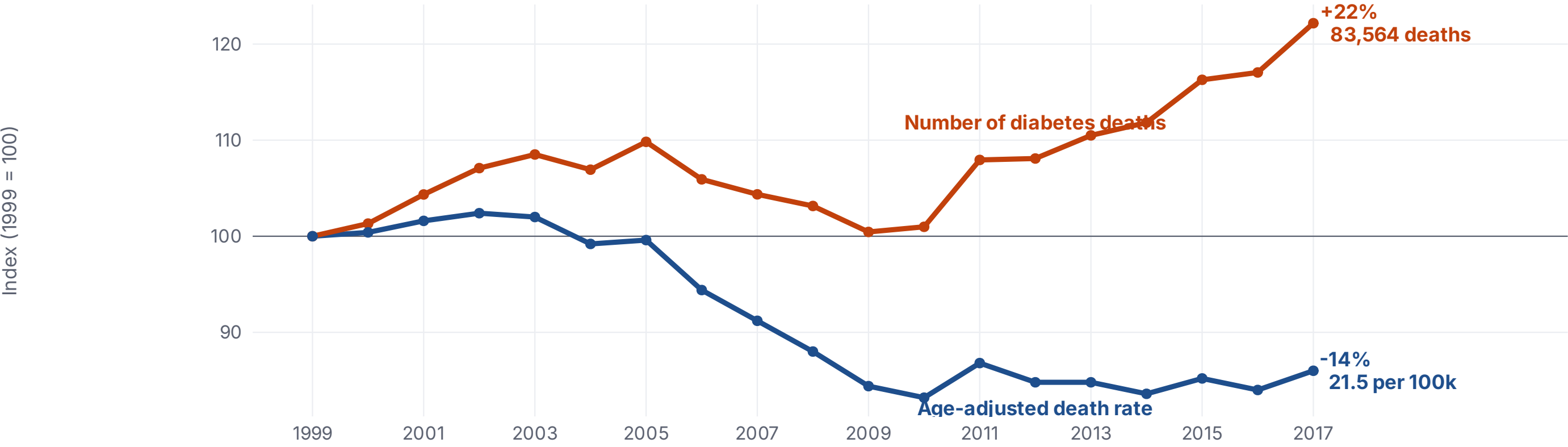
Highest 10 and lowest 5 states

The South and Appalachia lead; New England is lowest.



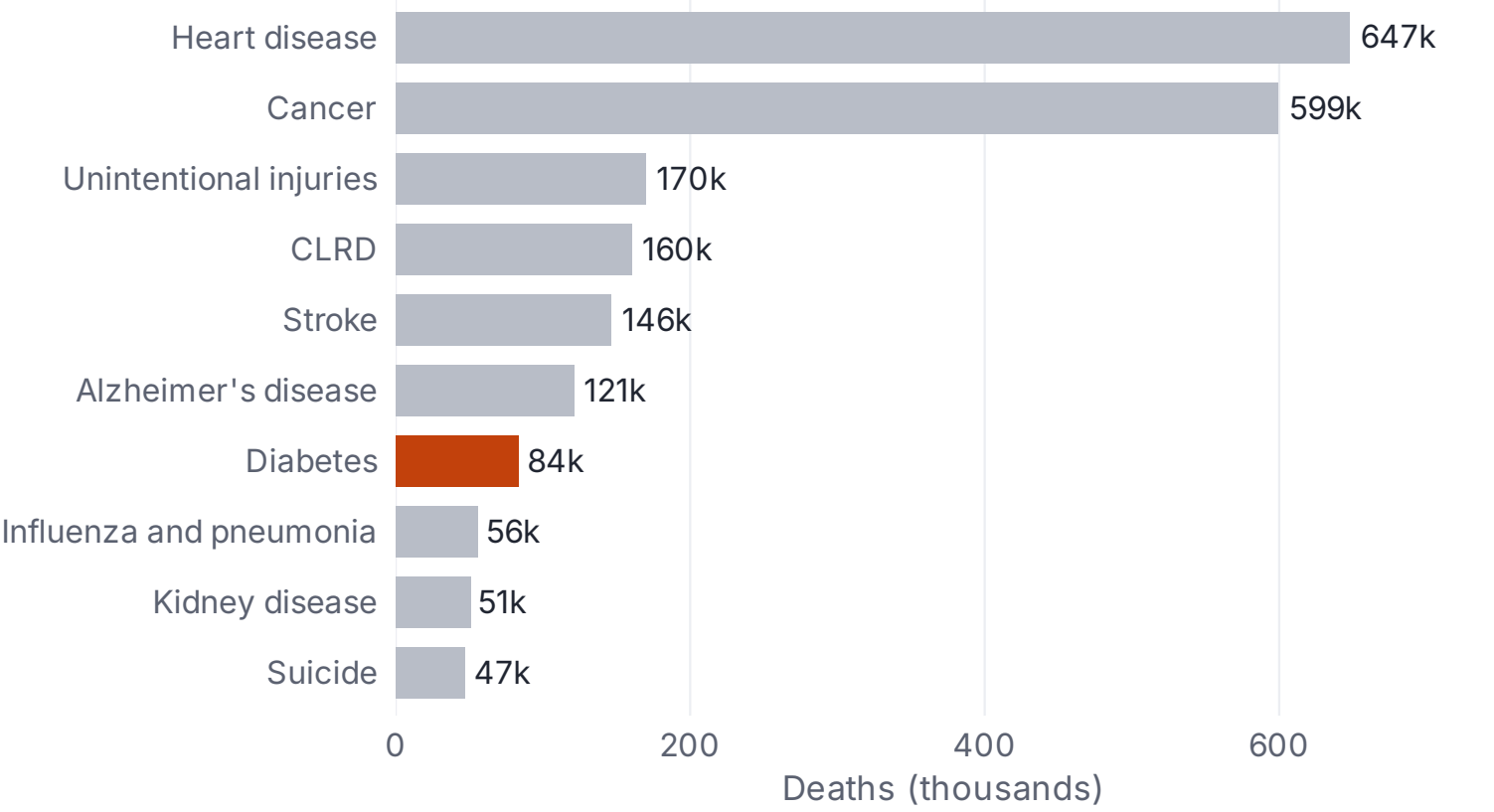
4. More people die of diabetes each year, yet each person's risk has fallen

US, 1999–2017, indexed to 1999 = 100. Deaths grow with a larger, older population (68,399 → 83,564); the age-adjusted rate removes that effect (25.0 → 21.5 per 100,000).



Diabetes ranks 7th

US leading causes, 2017



Why it matters

Diabetes deaths concentrate in the South and Appalachia, in the same states with high heart, lung and cancer death rates. Prevention aimed at shared risk factors (obesity, smoking, inactivity, access to care) can reduce several causes of death at once.

Audience and action

- People with diabetes or prediabetes: screening and early management.
- Health systems and state health departments: focus resources on the highest-rate states.
- Food industry and policymakers: reduce added sugar; support healthier food environments.

Method and limits

Pearson correlations across 51 states (50 + DC) on age-adjusted rates averaged 1999–2017; p-values from two-sided tests ($n = 51$). State-level (ecological) correlations do not prove that diabetes causes other deaths, and rates are not adjusted for income or smoking. 'All causes' rows were excluded to avoid double counting. Built entirely in R (ggplot2, patchwork).